IGCSE 0607 Practice Questions — E3.1

Graphs of functions

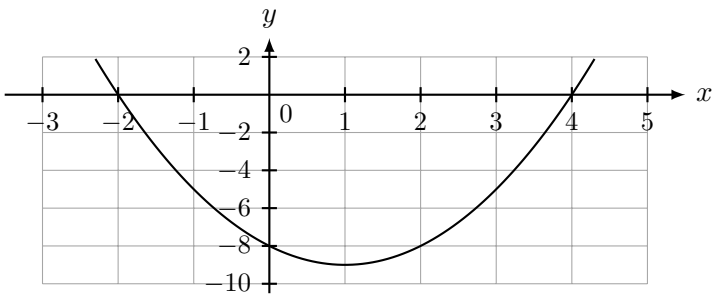
Total Questions: 3 — **Total Marks:** 16 Marks

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1 The diagram shows the graph of the function $y = x^2 + bx + c$.



(a) Write down the value of c.

$c =$ ..... **[1]**

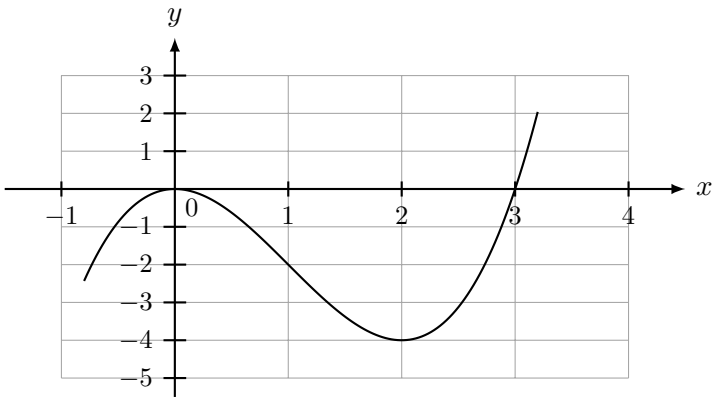
(b) Find the value of b.

$b =$ ..... **[2]**

(c) Write down the coordinates of the minimum point of the graph.

(..... , ...) **[2]**

2 The diagram shows the graph of $y = f(x)$ where $f(x) = ax^3 + bx^2$.
The graph passes through the origin $(0, 0)$, the point $(3, 0)$ and has a local minimum point at $(2, -4)$.



(a) Show that $3a + b = 0$.

[1]

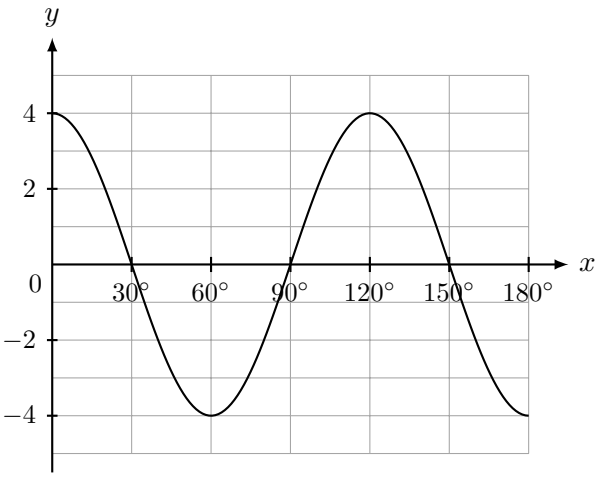
(b) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(c) Write down the range of values of k for which the equation $f(x) = k$ has exactly one solution.

..... **[2]**

3 The diagram shows a sketch of the graph of $f(x) = a\cos(bx)$ for $0° \leq x \leq 180°$.



(a) Write down the value of a.

$a =$ ..... **[1]**

(b) Find the value of b.

$b =$ ..... **[2]**

(c) Determine the number of solutions to the equation $a\cos(bx) = 1.5$ in the interval $0° \leq x \leq 360°$.

..... **[2]**